2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Learning VS Code + Learning GitHub.
Bansil Patel & Meet Patel: Learning VS Code + Exploring Dataset.
Harsh Mohile & Rachit Bhatt: Learning VS Code + Creating Branches for automation and deployment, and over Git Management.

2.2 z } µ Œ š u [• š • l •

- x Adapting VS Code as IDE for development and version control.
- x GitHub Management.
- x Exploring Dataset.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Understood basics of VS Code Extensions and GitHub.
Bansil Patel & Meet Patel: Applying NN and Model Evaluation through Summary.
Harsh Mohile: Worked on the quality assurance branch for applying test automation.
Rachit Bhatt: Worked on the production branch for launching the tested pieces of solution to the cloud.

3.2 z } µ Œ š progress

- x GitHub Management for Deployment and Test Automation.
- x Applying initial NN function and testing the results on the sample data.

4. The areas/tasks you could not make progress and/or complete as scheduled the difficulties encountered in this reporting week at individual level and group level

4.1 Individual project blockers

Sakshi: N/A
Bansil Patel: N/A
Harsh Mohile: N/A
Meet Patel: N/A
Rachit Bhatt: N/A

4.2 z } µ Œ š u [• ‰ Œ } i š o } l Œ •

- x Since the tasks of this scrum were quite smooth, luckily; there were no blockers in this week.

5. Tasks to be completed in next week at individual level and group level

5.1 Individual tasks

Sakshi: Research on CI/CD Pipelining for MLOps.

Bansil Patel & Meet Patel: Applying various models initially on the dataset.

Harsh Mohile & Rachit Bhatt: Research on Docker for Deployment.

5.2 z } µ Œ š tasks •

- x Exploring and understanding the dataset.
- x Research on Docker for Deployment.

6. Include the tasks from your sprint planning (Github/Zenhub) for the present period.

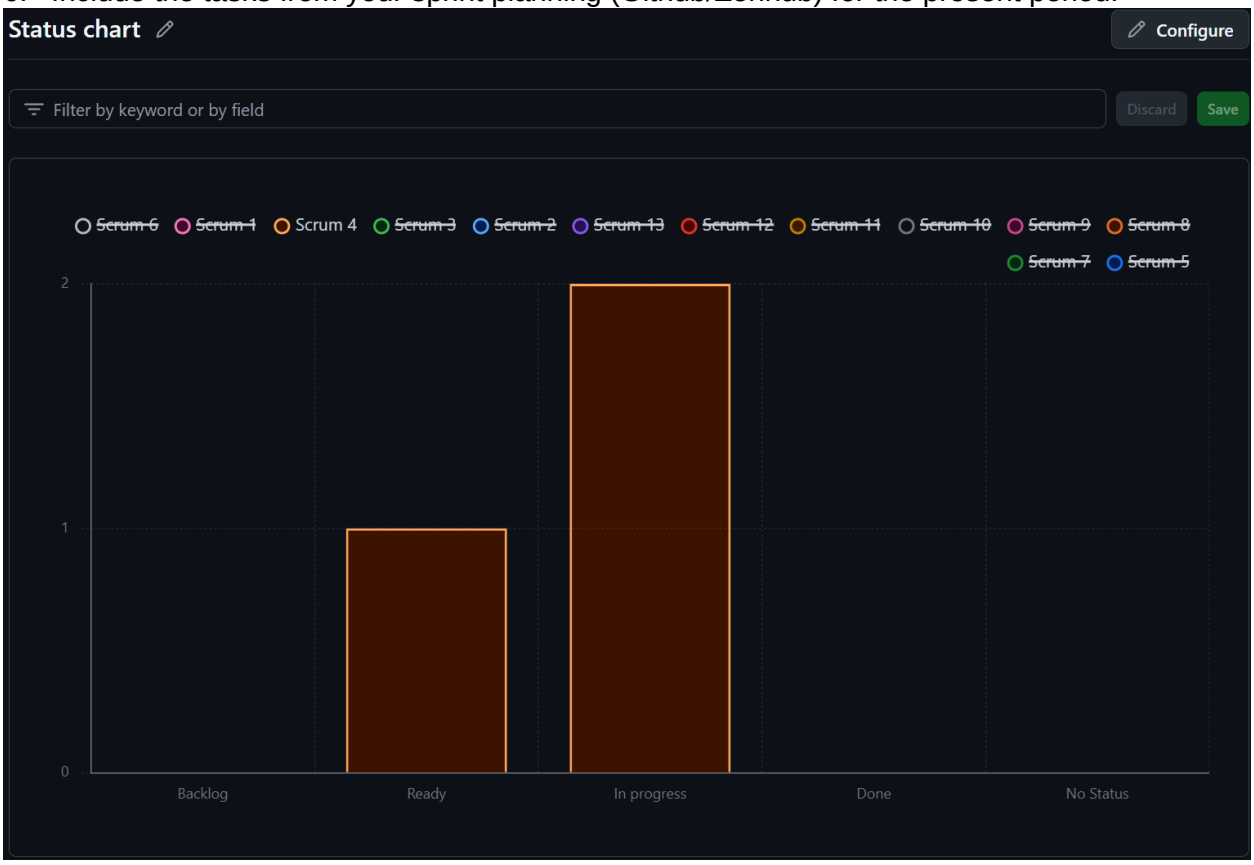

Figure1: Status Chart of GitHub Issues in Milestone

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

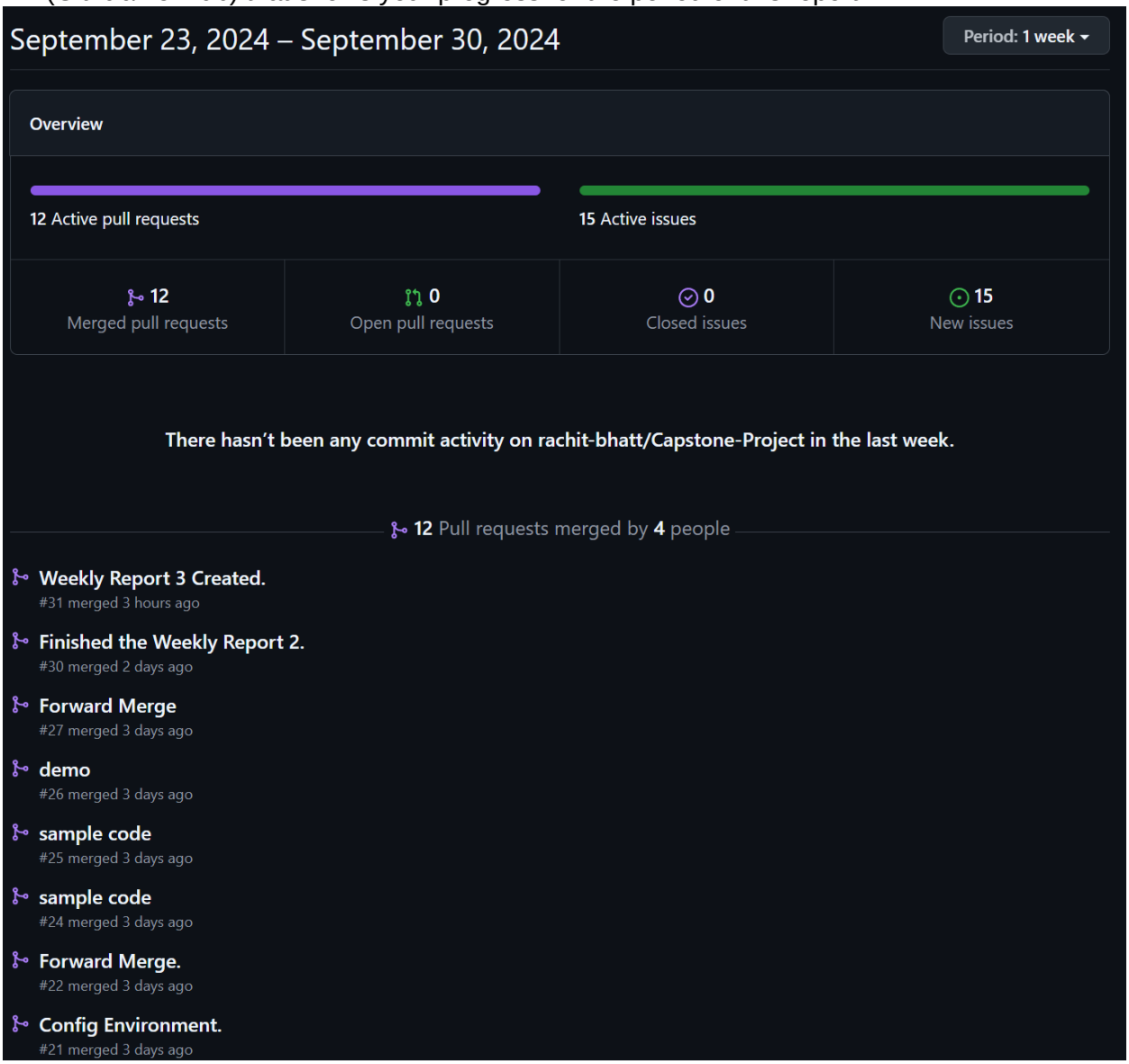

Figure2: Contributors in the Week

8. Include a note and address to your project Github with list of codes uploaded/updated on Github in this reporting week.

GitHub: https://www.github.com/rachit-bhatt/CapstoneProject

```
epochs=5

# Train the model
history_eff = model_eff.fit(
    train_ds,
    epochs=epochs,
    validation_data=valid_ds,
    verbose=1,
)

# Save training and validation histories for later analysis
all_train_histories = [history_eff.history['accuracy']]
all_val_histories = [history_eff.history['val_accuracy']]
```

```
Epoch 1/5
35/35 ━━━━━━━━━━━━━━━━━━━━ 188s 5s/step - AUC: 0.6027 - accuracy: 0.3963 - f1_score: 0.1340 - loss: 10.3684 - val_AUC: 0.5114 - val_accuracy: 0.3745 - val_f1_score: 0.3554 - val_loss: 396.7017
Epoch 2/5
35/35 ━━━━━━━━━━━━━━━━━━━━ 250s 7s/step - AUC: 0.5140 - accuracy: 0.3471 - f1_score: 0.0000e+00 - loss: 5.8994 - val_AUC: 0.4966 - val_accuracy: 0.3636 - val_f1_score: 0.2646 - val_loss: 52.6579
Epoch 3/5
35/35 ━━━━━━━━━━━━━━━━━━━━ 209s 6s/step - AUC: 0.4978 - accuracy: 0.3455 - f1_score: 0.0000e+00 - loss: 5.2170 - val_AUC: 0.5377 - val_accuracy: 0.3818 - val_f1_score: 0.1321 - val_loss: 15.6876
Epoch 4/5
35/35 ━━━━━━━━━━━━━━━━━━━━ 157s 4s/step - AUC: 0.5134 - accuracy: 0.3648 - f1_score: 0.0000e+00 - loss: 4.7851 - val_AUC: 0.5799 - val_accuracy: 0.3927 - val_f1_score: 0.0608 - val_loss: 8.3147
Epoch 5/5
35/35 ━━━━━━━━━━━━━━━━━━━━ 159s 5s/step - AUC: 0.4948 - accuracy: 0.3574 - f1_score: 0.0000e+00 - loss: 4.4633 - val_AUC: 0.5314 - val_accuracy: 0.3236 - val_f1_score: 0.0583 - val_loss: 8.4688
```

Figure3: Applying NN on a Sample Dataset

```
img_shape = (img_size[0], img_size[1], 3)

base_model = DenseNet169(include_top=False, weights="imagenet", input_shape=img_shape, pooling='max')
base_model.trainable = True
x = base_model.output
x = BatchNormalization(axis=-1, momentum=0.99, epsilon=0.001)(x)
x = Dense(256, kernel_regularizer=regularizers.l2(0.016),
          activity_regularizer=regularizers.l1(0.006),
          bias_regularizer=regularizers.l1(0.006), activation='relu')(x)
x = Dropout(rate=.4, seed=123)(x)
output = Dense(class_count, activation='softmax')(x)
model_eff = Model(inputs=base_model.input, outputs=output)
model_eff.compile(Adamax(learning_rate=lr), loss='categorical_crossentropy',
                  metrics=['accuracy', 'AUC', F1Score()])

model_eff.summary()
```

```
Model: "functional"
```

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
zero_padding2d (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer[0][0]
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,408	zero_padding2d[0…
conv1_bn (BatchNormalizatio…	(None, 112, 112, 64)	256	conv1_conv[0][0]

Figure4: Model Evaluation Summary

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) Bansil Patel (C0912873) – TL Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 5
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Research on CI/CD Pipelining for ML-Ops.
Bansil Patel & Meet Patel: Applying various models initially on the dataset.
Harsh Mohile & Rachit Bhatt: Research on Docker for Deployment.

2.2 Your team's tasks

- Exploring and understanding the dataset.
- Research on Docker for Deployment.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Studied the theoretical concepts of CI/CD.
Bansil Patel: Applied regular NN with Model Evaluation and attempted in accessing the annotated data in the JSON files.
Harsh Mohile & Rachit Bhatt: Collaborated to create a demo project using Docker.

3.2 Your team's progress

- Worked on understanding Docker by creating a demo project; GitHub - Docker.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A
Bansil Patel & Meet Patel: `AttributeError` exception generated when trying to access the JSON files.
Harsh Mohile & Rachit Bhatt: Docker installation and implementation on Windows platform was a little challenging when finding proper content for research and development.

4.2 Your team's project blockers

- Faced a progress blocker in the form of exception when trailing with JSON files; seems like an OS-level error due to some path or incorrect data provided to the library.

5. Tasks to be completed in next week at individual level and group level

5.1 Individual tasks

Sakshi: Implementation CI/CD on a demo project.

Bansil Patel | Harsh Mohile | Meet Patel: Resolve the exception and implement multi-model approach.

Rachit Bhatt: Start Implementing CI/CD using Docker through GitHub.

5.2 Your team's tasks

- Work on multi-model setup.
- Implement CI/CD using Docker with GitHub.

6. Include the tasks from your sprint planning (Github/Zenhub) for the present period.

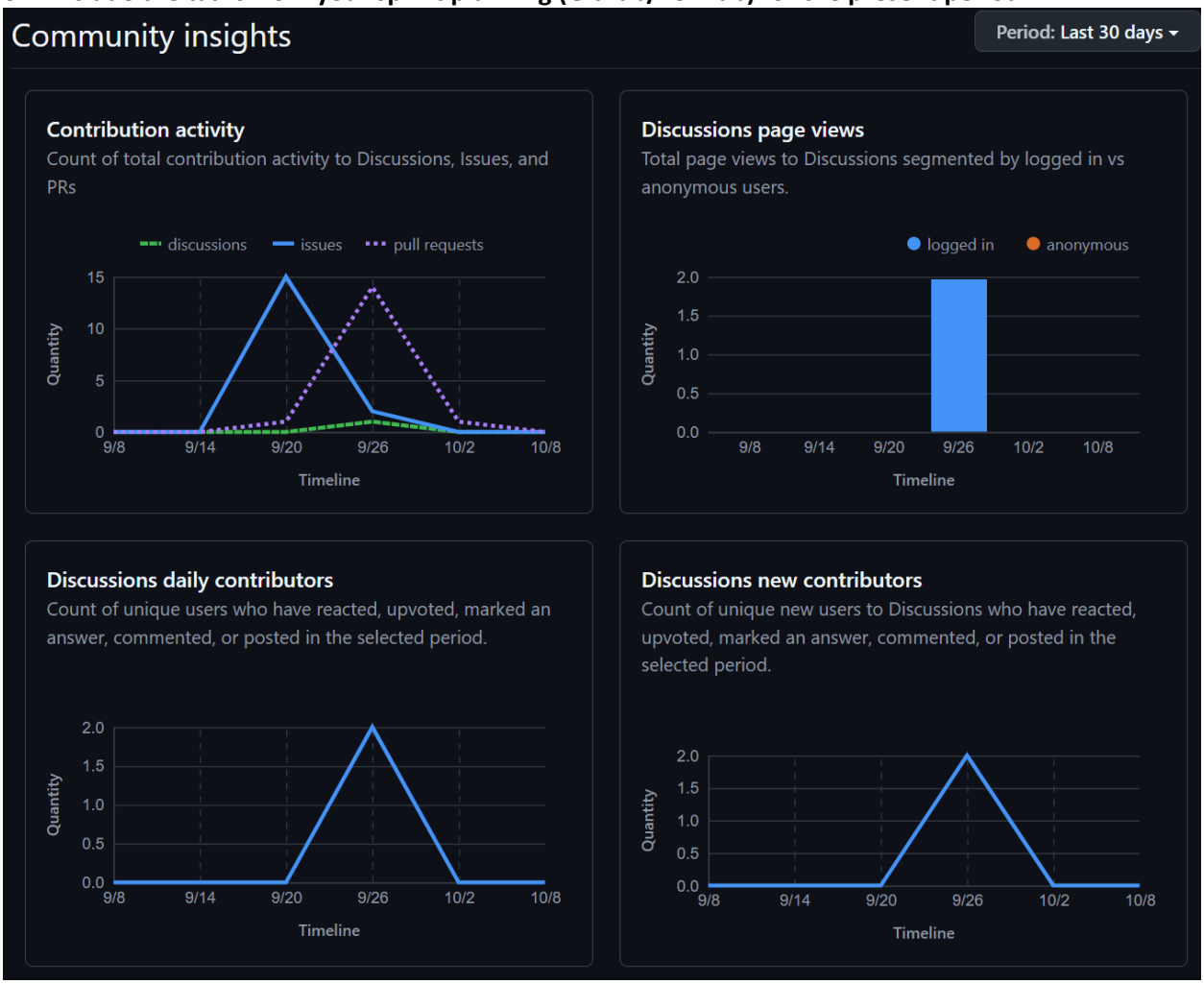

Figure 1: Weekly Activities

- In this week, there were 15 issues resolved with numerous PRs, contributing in development.

7. **Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.**

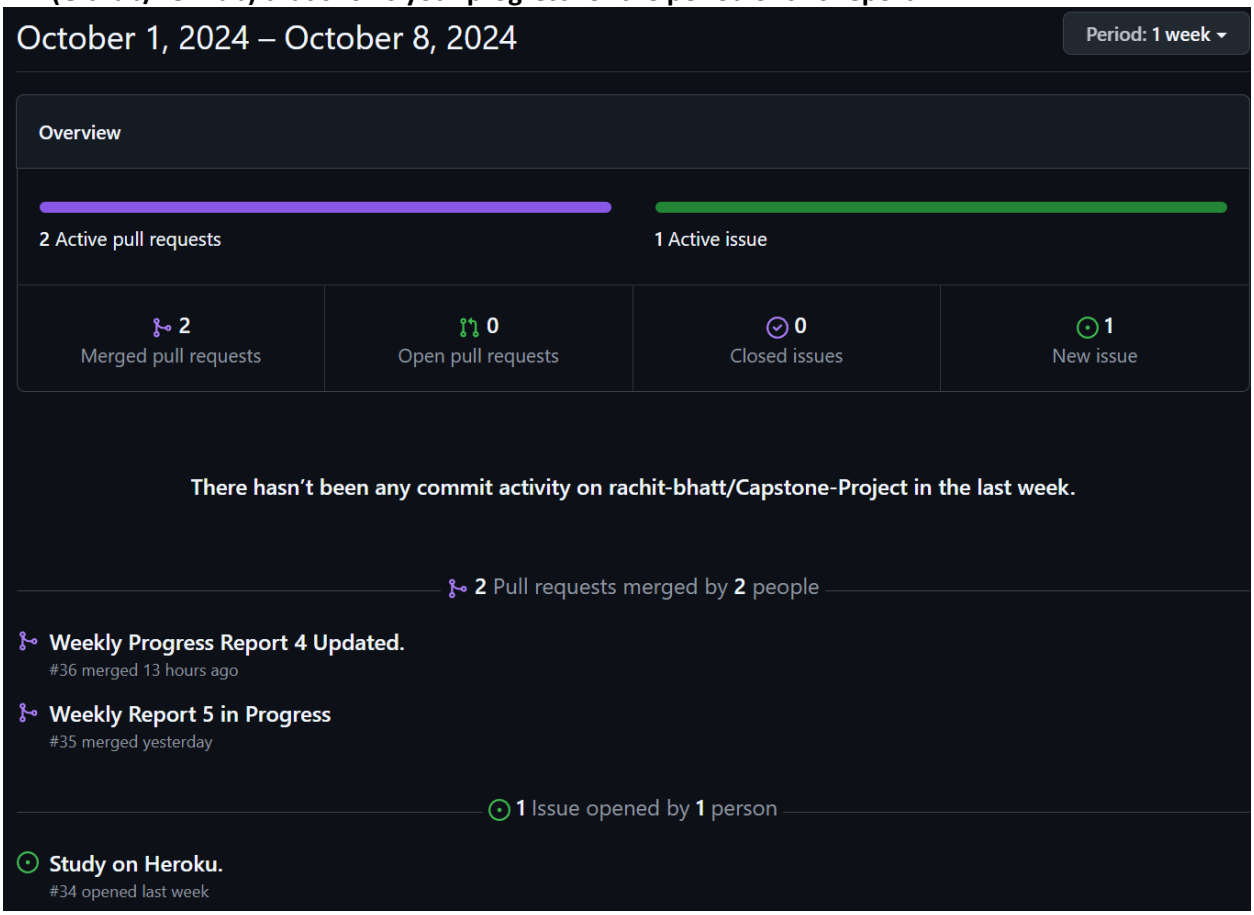

Figure 2: Contributors in the Week

- This week, there were two pull requests generated and merged that included phase 1 of our project and documentation.
- The one active issue we dealt with was the exception generated as shown in the *Figure 3*, took a lot of time and decided to shift it to the next scrum.

8. **Include a note and address to your project Github with list of codes uploaded/updated on Github in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

```
            predict_and_display_image('uploaded_image.jpg', model_eff)

# Attach the function to the upload button
upload_btn.observe(on_upload_change, names='value')

# Display the upload button and output widget
display(upload_btn, output)
```

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"428ebbf54e264659a871ca536e14de18","version_major":2,"version_minor":0}

Could not render content for 'application/vnd.jupyter.widget-view+json'
{"model_id":"4f7b00a88f9c4deb8036a8a75e08fa71","version_major":2,"version_minor":0}

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
File c:\Users\pbans\OneDrive\Desktop\CD2\.conda\Lib\site-packages\ipywidgets\widgets\widget.py:773, in Widget._handle_msg(self, msg)
    771         if 'buffer_paths' in data:
    772             _put_buffers(state, data['buffer_paths'], msg['buffers'])
--> 773         self.set_state(state)
    775 # Handle a state request.
    776 elif method == 'request_state':

File c:\Users\pbans\OneDrive\Desktop\CD2\.conda\Lib\site-packages\ipywidgets\widgets\widget.py:650, in Widget.set_state(self, sync_data)
    645         self._send(msg, buffers=echo_buffers)
    647 # The order of these context managers is important. Properties must
    648 # be locked when the hold_trait_notification context manager is
    649 # released and notifications are fired.
--> 650 with self._lock_property(**sync_data), self.hold_trait_notifications():
    651     for name in sync_data:
    652         if name in self.keys:

File c:\Users\pbans\OneDrive\Desktop\CD2\.conda\Lib\contextlib.py:144, in _GeneratorContextManager.__exit__(self, typ, value, traceback)
    142 if typ is None:
    143     try:
--> 144         next(self.gen)
    145     except StopIteration:
    146         return False
...
     41         # Save the uploaded image to a temporary path
     42         with open('uploaded_image.jpg', 'wb') as f:
     43             f.write(file_info['content'])

AttributeError: 'tuple' object has no attribute 'items'
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

Figure 3: Exception – Progress Blocker

- The exception when dealing with the image by the end user.
- Expecting it to be resolved in the next scrum.

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) – TL Rachit Bhatt (C0902810)
Reporting Week	Week 6
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Implementation CI/CD on a demo project.

Bansil Patel | Harsh Mohile | Meet Patel: Resolve the exception and implement a multi-model approach.

Rachit Bhatt: Start Implementing CI/CD using Docker through GitHub.

2.2 Your team's tasks

- Work on multi-model setup.
- Implement CI/CD using Docker with GitHub.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Implemented CI/CD demonstration project.

Bansil Patel | Harsh Mohile | Meet Patel: Resolved exception and applied other models.

Rachit Bhatt: Initiated docker setup in local system.

3.2 Your team's progress

- Worked on understanding Docker by creating a demo project; GitHub - Docker.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A

Bansil Patel: N/A

Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: Docker installation and implementation on Windows platform was a little challenging when finding proper content for research and development.

4.2 Your team's project blockers

- Installation of docker blocked the progress speed.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Research on docker.

Bansil Patel & Meet Patel: Analyze model outcomes and implement metrics.

Harsh Mohile & Rachit Bhatt: Demonstrate GitHub Actions.

5.2 Your team's tasks

- Analyze model performance and implement metrics.
- Understand working of GitHub Actions.

6. Include the tasks from your sprint planning (Github/Zenhub) for the present period.

Figure 1: Weekly Activities

- In this week, there were 3 contributions made with one PRs, contributing in development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

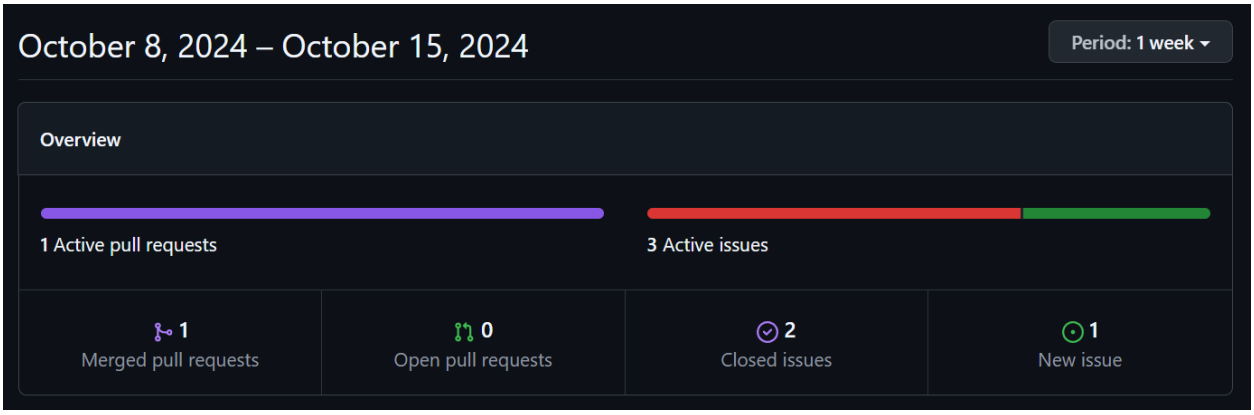

Figure 2: Contributors in the Week

- This week, due to researching topics, we had less PRs compared to previous week.

8. Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 7
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Research on docker.
Bansil Patel & Meet Patel: Analyze model outcomes and implement metrics.
Harsh Mohile & Rachit Bhatt: Demonstrate GitHub Actions.

2.2 Your team's tasks

- Analyze model performance and implement metrics.
- Understand working of GitHub Actions.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Workflow of docker and its importance.
Bansil Patel & Meet Patel: Analyzed model outcomes and initiated working on metrics.
Harsh Mohile & Rachit Bhatt: Studied GitHub Actions.

3.2 Your team's progress

- Worked on understanding GitHub Actions by creating a demo project; GitHub - Actions.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A
Bansil Patel: N/A
Meet Patel: N/A
Harsh Mohile & Rachit Bhatt: Improving GitHub deployment for each yml created and overhead of failures.

4.2 Your team's project blockers

- Installation of docker blocked the progress speed.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Install docker and work on creating containers.
Bansil Patel & Meet Patel: Apply alternative models or initiate hyperparameter tuning.

Harsh Mohile & Rachit Bhatt: Create a demo project on GitHub Actions.

5.2 Your team's tasks

- Apply new models and hyperparameter tuning.
- Demonstrated a project using GitHub Actions.

6. Include the tasks from your sprint planning (Github/Zenhub) for the present period.

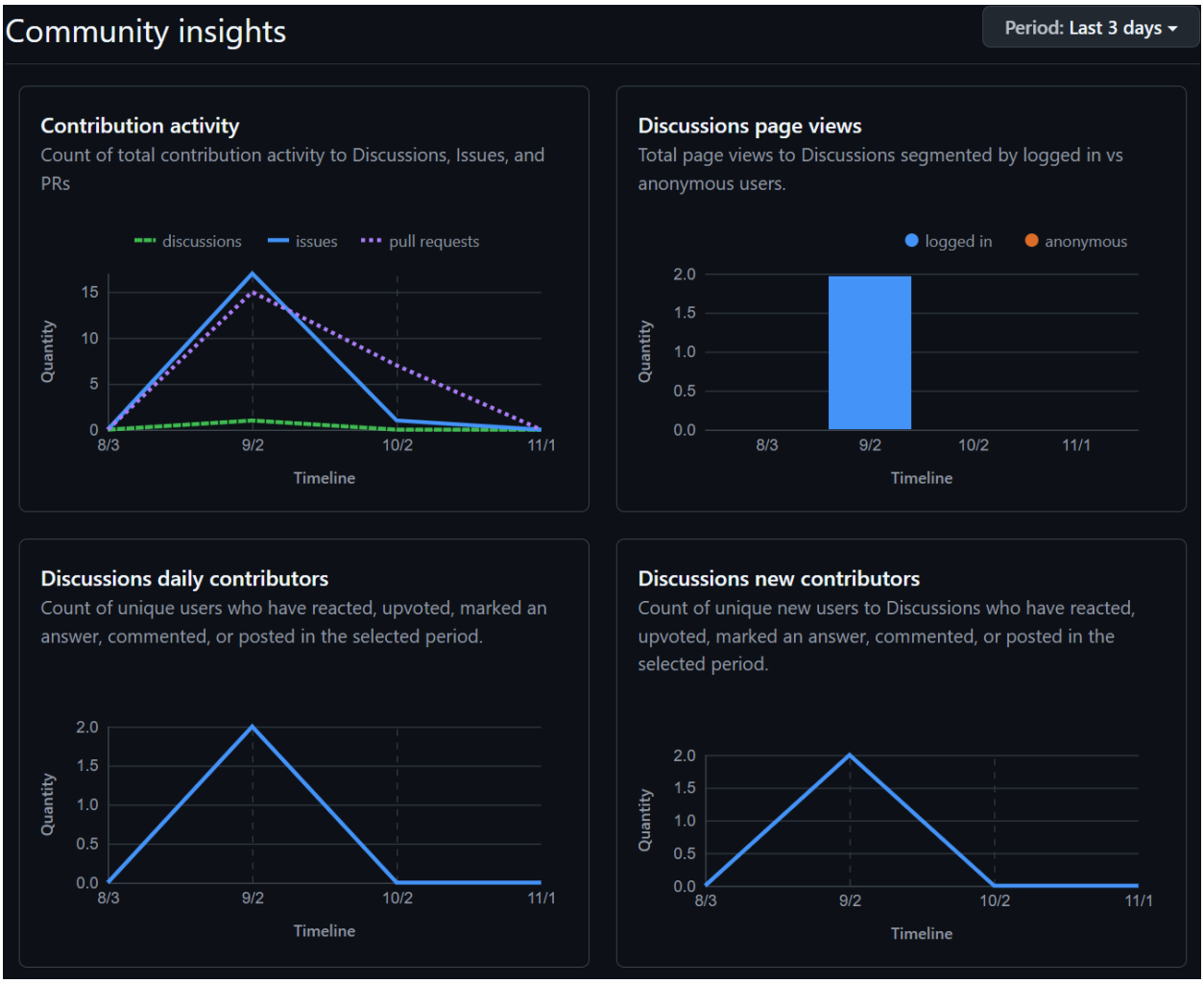

Figure 1: Weekly Activities

- In this week, there were 17 contributions made with 15 PRs, contributing in development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

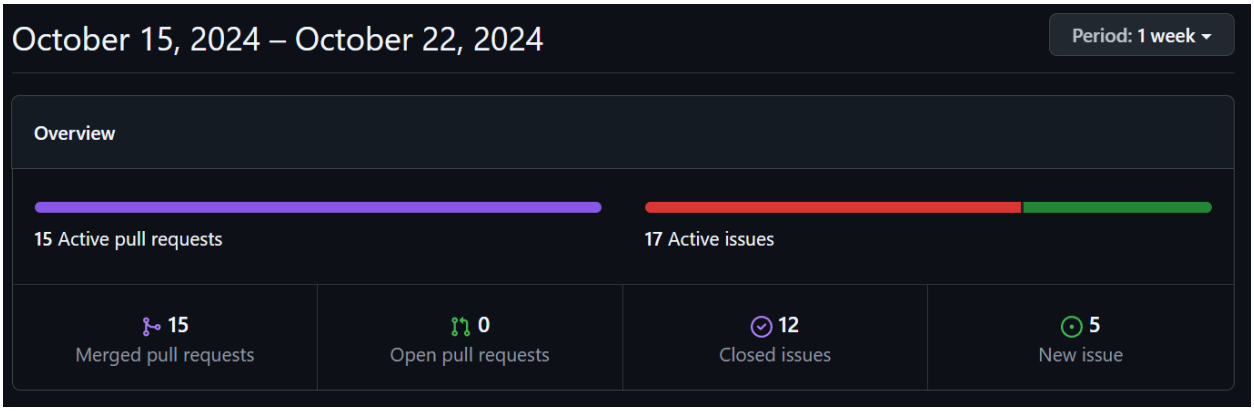

Figure 2: Contributors in the Week

- This week, we had quite a lot of PRs compared to previous week, demonstrating team hard work and dedication.

8. Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810) – TL
Reporting Week	Week 8
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Install docker and work on creating containers.
Bansil Patel & Meet Patel: Apply alternative models or initiate hyperparameter tuning.
Harsh Mohile & Rachit Bhatt: Create a demo project on GitHub Actions.

2.2 Your team's tasks

- Apply new models and hyperparameter tuning.
- Demonstrated a project using GitHub Actions.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Installation of docker and creating containers for sample.
Bansil Patel & Meet Patel: Apply alternative models.
Harsh Mohile & Rachit Bhatt: Demonstrated a sample project using GitHub Actions.

3.2 Your team's progress

- Worked on understanding GitHub Actions by creating a demo project; GitHub - Actions.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A
Bansil Patel: N/A
Meet Patel: N/A
Harsh Mohile & Rachit Bhatt: CI/CD for multiple branches on GitHub when tweaking automative docker deployment script.

4.2 Your team's project blockers

- This week, our team did not face as such blockers.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Understand need for GitHub Actions in our project.

Bansil Patel & Meet Patel: Analyze model metrics and compare them to proceed for hyperparameter tuning.

Harsh Mohile & Rachit Bhatt: Demonstrate automative docker deployment for CI/CD on GitHub through GitHub Actions.

5.2 Your team's tasks

- Analyze model metrics for hyperparameter tuning.
- Demonstrate automative docker deployment script.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

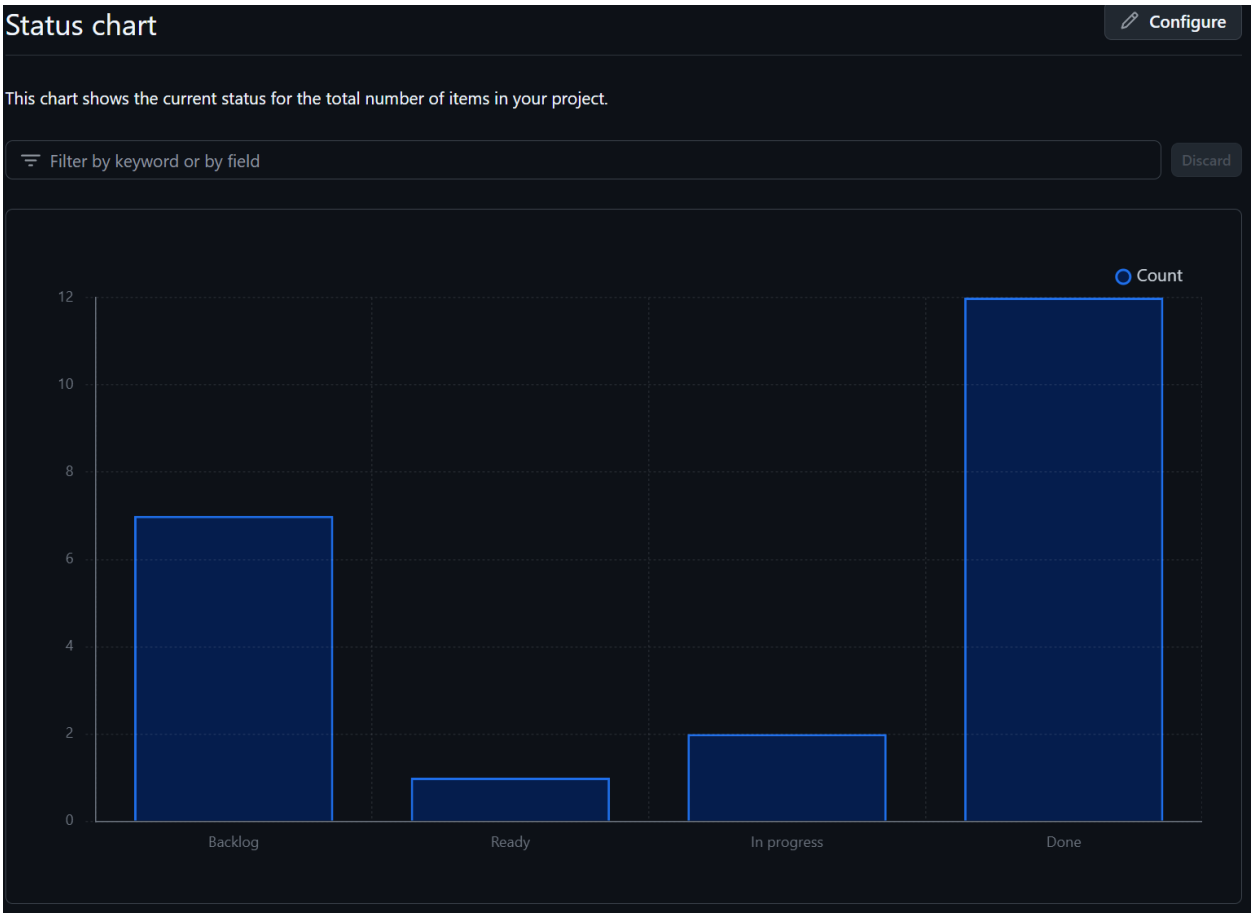

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. **Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.**

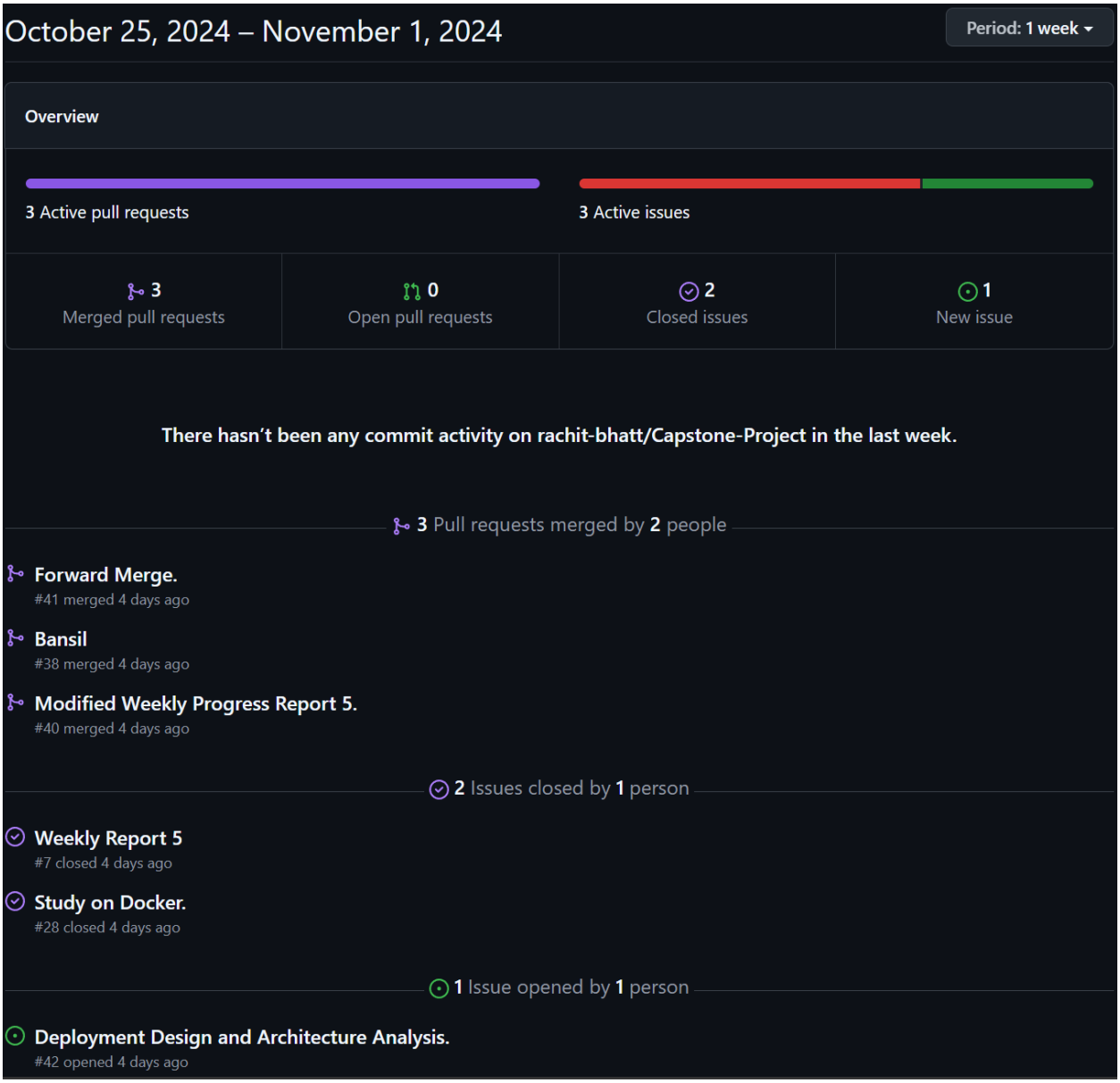

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 9
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Completed Docker Installation and successfully created containers for the project. Explored the use of Docker Compose for handling multiple containers.

Bansil Patel & Meet Patel: Conducted performance evaluations of different models and began the process of tuning hyperparameters on the selected model.

Harsh Mohile & Rachit Bhatt: Showcased the setup and execution of CI/CD pipelines using Github Actions and Docker. Worked on optimizing Github Actions workflows for automatic deployment.

2.2 Your team's tasks

- **Model Application:** Progressed in evaluating alternative models and started the hyperparameter tuning process based on the model performance comparison.
- **CI/Cd Pipeline Demonstration:** Successfully demonstrated how Github Actions can be used to automate Dcker deployments, streamlining the CI/CD process for the project.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Set up Docker containers and explored Docker Compose for multi-container management.

Bansil Patel & Meet Patel: Evaluated model performance (accuracy, speed) and started hyperparameter tuning on the selected model.

Harsh Mohile & Rachit Bhatt: Demonstrated automated deployment using Github Actions and Docker, optimizing windows.

3.2 Your team's progress

- Configured Github Actions to automate Docker deployment and set up a CI/CD pipeline.
- Completed model evaluation and are refining the selected model with hyperparameter tuning.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A

Bansil Patel: N/A

Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: Faced challenges while setting up CI/Cd workflows for multiple branches in Github, particularly in aligning Docker deployment with branch specific configurations.

4.2 Your team's project blockers

- While no major blockers were encountered this week, the team experienced some delays in configuring the CI/CD pipeline for multiple branches, particularly when adjusting the Docker deployment scripts.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Investigate how GitHub Actions can be integrated into the project for automating testing and deployment.

Bansil Patel & Meet Patel: Continue analyzing the model metrics and complete the hyperparameter tuning to optimize performance.

Harsh Mohile & Rachit Bhatt: Demonstrate a fully automated Docker deployment process for multiple environments (e.g., development, staging, production) using GitHub Actions.

5.2 Your team's tasks

- Finalize the analysis of the model's performance metrics and proceed with hyperparameter optimization.
- Continue enhancing the Docker deployment workflow using GitHub Actions and ensure it works seamlessly across different environments.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

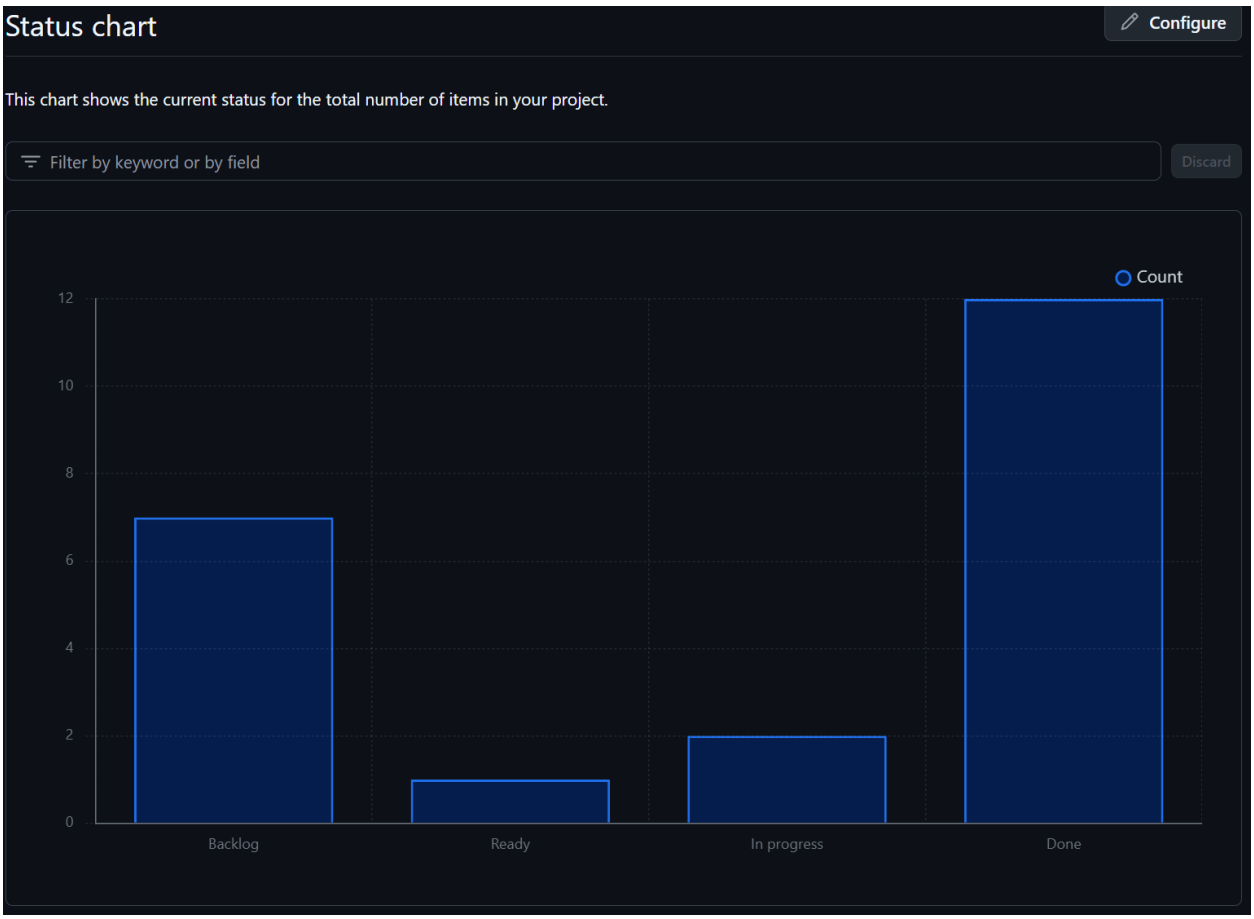

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

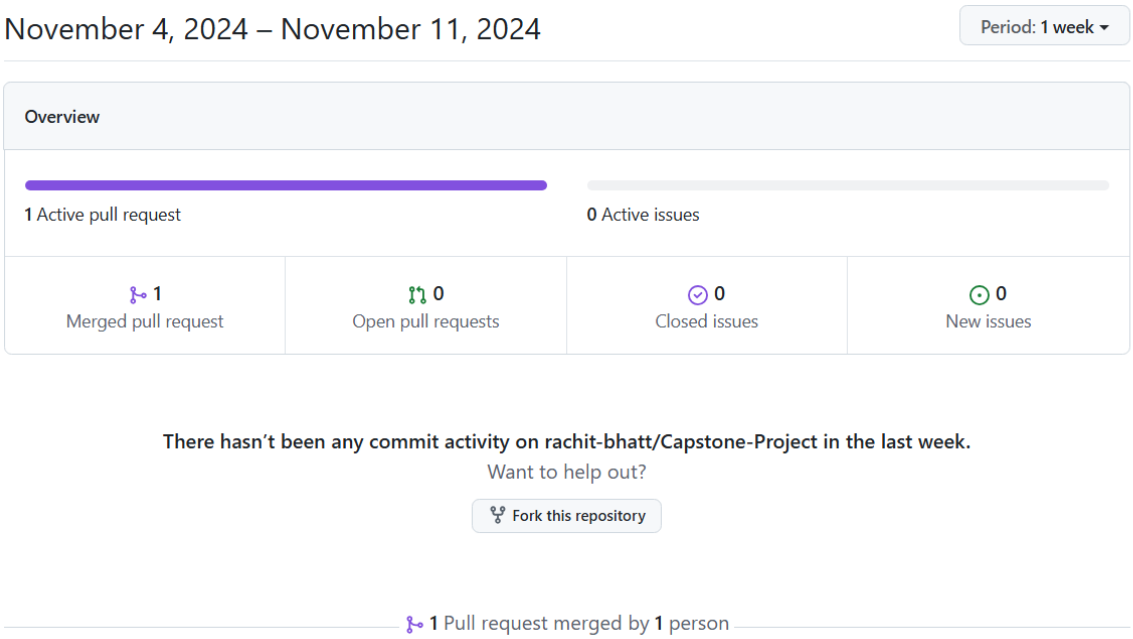

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 10
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Investigate how GitHub Actions can be integrated into the project for automating testing and deployment.

Bansil Patel & Meet Patel: Continue analyzing the model metrics and complete the hyperparameter tuning to optimize performance.

Harsh Mohile & Rachit Bhatt: Demonstrate a fully automated Docker deployment process for multiple environments (e.g., development, staging, production) using GitHub Actions.

2.2 Your team's tasks

- Finalize the analysis of the model's performance metrics and proceed with hyperparameter optimization.
- Continue enhancing the Docker deployment workflow using GitHub Actions and ensure it works seamlessly across different environments.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Set up Docker containers and explored Docker Compose for multi-container management.

Bansil Patel & Meet Patel: Evaluated model performance (accuracy, speed) and started hyperparameter tuning on the selected model.

Harsh Mohile & Rachit Bhatt: Demonstrated automated deployment using GitHub Actions and Docker, optimizing windows.

3.2 Your team's progress

- Configured GitHub Actions to automate Docker deployment and set up a CI/CD pipeline.
- Completed model evaluation and refining the selected model with hyperparameter tuning.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: N/A

Bansil Patel: N/A

Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: Faced challenges while setting up CI/CD workflows for multiple branches in GitHub, particularly in aligning Docker deployment with branch specific configurations.

4.2 Your team's project blockers

- While no major blockers were encountered this week, the team experienced some delays in configuring the CI/CD pipeline for multiple branches, particularly when adjusting the Docker deployment scripts.

5. Tasks to be completed in the next week at individual level and group level

5.1 Individual tasks

Sakshi: Investigate how GitHub Actions can be integrated into the project for automating testing and deployment.

Bansil Patel & Meet Patel: Continue analyzing the model metrics and complete the hyperparameter tuning to optimize performance.

Harsh Mohile & Rachit Bhatt: Researching Deployment UI and Backend.

5.2 Your team's tasks

- Finalize the analysis of the model's performance metrics and proceed with hyperparameter optimization.
- Researching Deployment UI and Backend.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

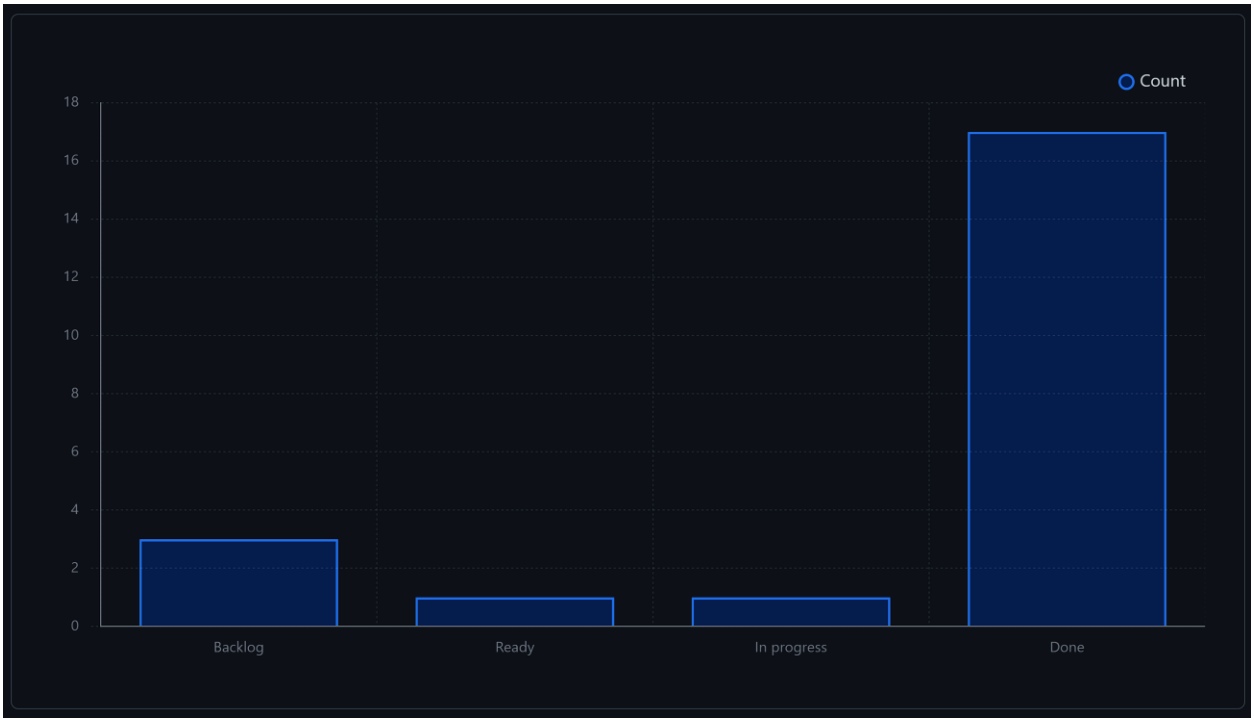

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

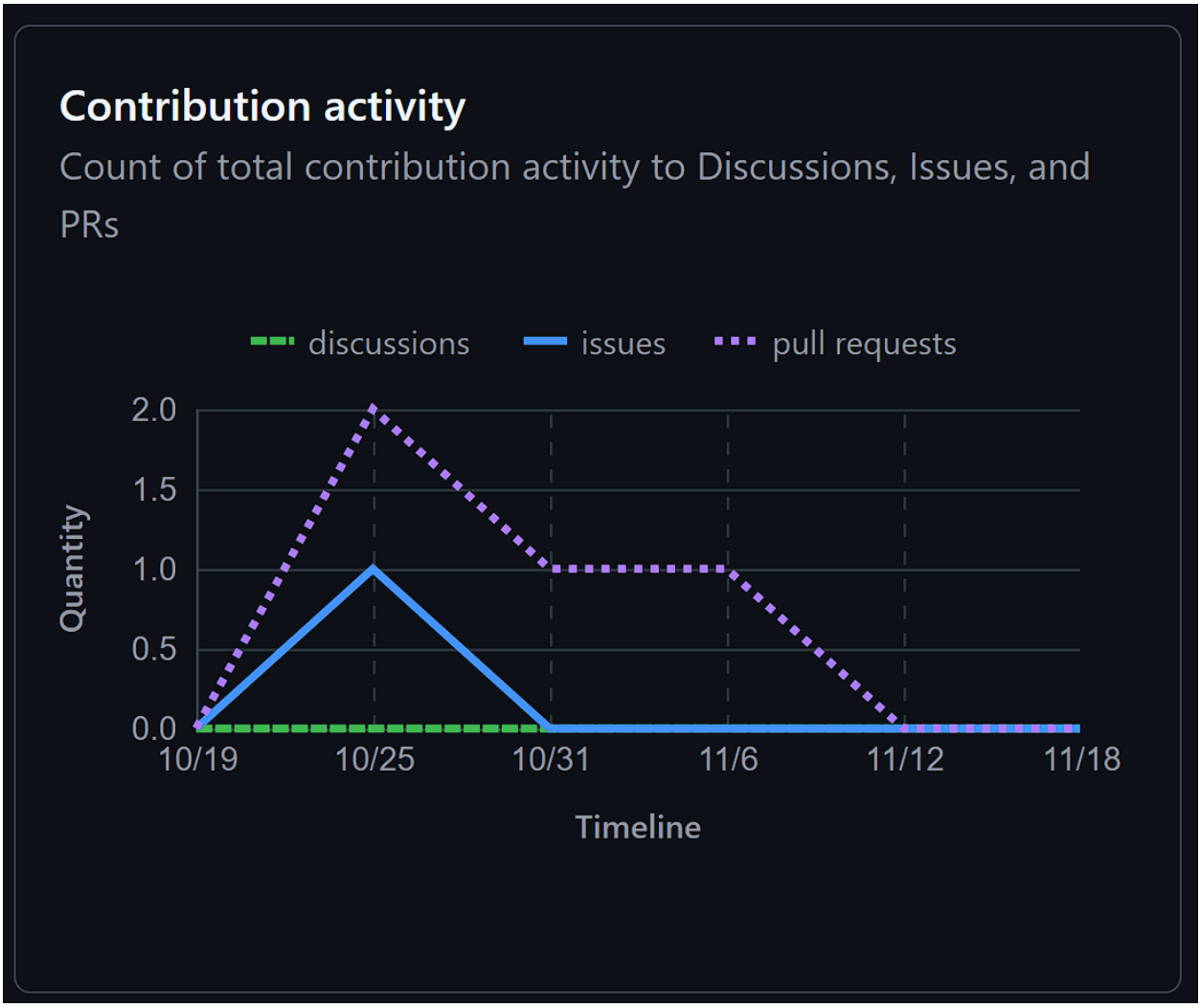

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 12
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Enhance the GitHub Actions pipeline by adding real-time monitoring and alerting features to track deployment status.

Bansil Patel & Meet Patel: Plan to expand testing with more diverse datasets, focusing on rare car damage types for improved model generalization**.**

Harsh Mohile & Rachit Bhatt: Further optimize the Deployment UI to improve user experience and make it more intuitive for end users.

2.2 Your team's tasks

- Model Enhancements: Plan to explore advanced model architectures to improve accuracy, aiming for higher performance in detecting complex car damages.
- Deployment Pipeline: Add real-time monitoring to the deployment pipeline, enabling better tracking of deployment success and quick rollback if needed.
- UI Optimization: Improve the Deployment UI with new features such as interactive visualizations of model predictions and user feedback integration.
- Scalability: Focus on optimizing the model's performance for large-scale data processing to handle a wider variety of car images in real-world applications.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Resolved repository configuration issues to ensure consistent deployments.

Bansil Patel & Meet Patel: Validated the model's robustness by testing on multiple datasets, confirming its generalization to different car damage types.

Harsh Mohile & Rachit Bhatt: Conducted extensive testing across staging and production environments, ensuring reliable performance.

3.2 Your team's progress

- Finalized the automated deployment pipeline with GitHub Actions, ensuring smooth deployments across all environments. Fully integrated the UI with the backend, ensuring seamless deployment in staging and production.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: Minor integration issues were resolved by refining repository configurations.
Bansil Patel & Meet Patel: Delays in backend-UI communication but resolved during testing.
Harsh Mohile & Rachit Bhatt: N/A.

4.2Your team's project blockers

- Multi-branch setup caused delays, resolved with configuration adjustments.
- Minor adjustments to align the UI with production configurations.

5. Tasks to be completed in the next week at the individual level and group level

5.1 Individual tasks

Sakshi, Bansil Patel & Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: N/A

5.2 Your team's tasks

The project has been completed successfully.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

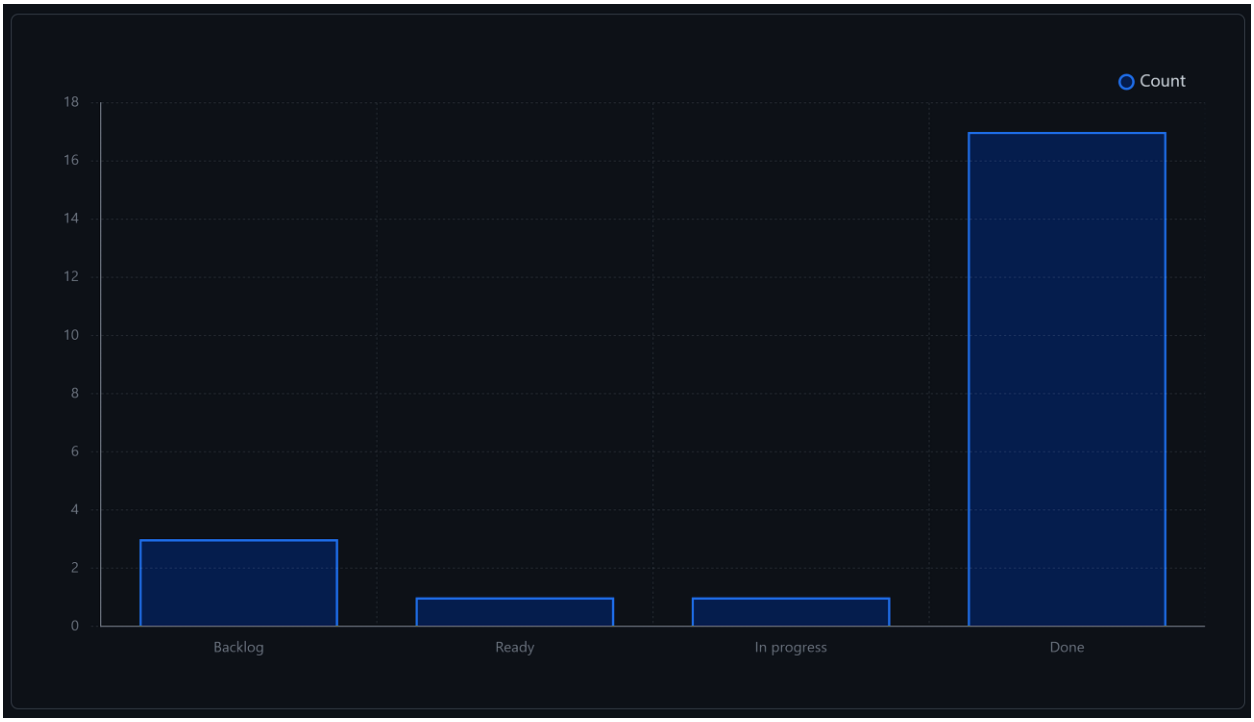

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.

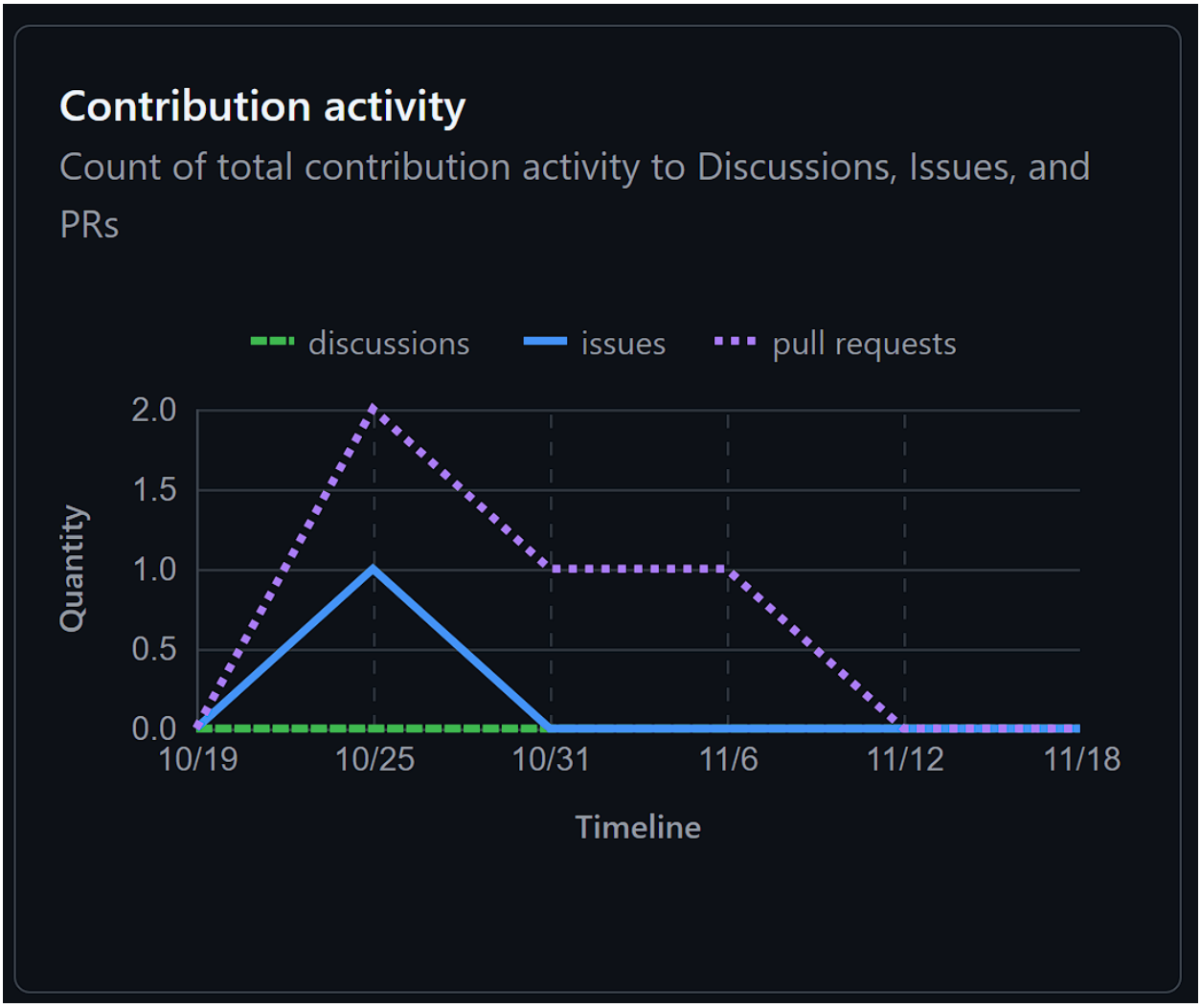

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project

Capstone Project Weekly Progress Report

Semester	Fall-2024 Semester 3rd
Course Code	AML-3406
Section	Section 1
Project Title	Car Damage Detection
Group Name	Group B
Student names/Student IDs	Sakshi (C0908000) – TL Bansil Patel (C0912873) Harsh Mohile (C0912872) Meet Patel (C0910378) Rachit Bhatt (C0902810)
Reporting Week	Week 12
Faculty Supervisor	William Pourmajidi

1. Grade the level of collaboration from different aspects between team members: (use: good, medium, below expectation)

Sakshi	Bansil Patel	Harsh Mohile	Meet Patel	Rachit Bhatt
Good	**Good**	**Good**	**Good**	**Good**

2. Tasks outlined in the previous Weekly Progress Report for this reporting week at the individual level and group level

2.1 Individual tasks (each team member)

Sakshi: Enhance the GitHub Actions pipeline by adding real-time monitoring and alerting features to track deployment status.

Bansil Patel & Meet Patel: Plan to expand testing with more diverse datasets, focusing on rare car damage types for improved model generalization**.**

Harsh Mohile & Rachit Bhatt: Further optimize the Deployment UI to improve user experience and make it more intuitive for end users.

2.2 Your team's tasks

- Model Enhancements: Plan to explore advanced model architectures to improve accuracy, aiming for higher performance in detecting complex car damages.
- Deployment Pipeline: Add real-time monitoring to the deployment pipeline, enabling better tracking of deployment success and quick rollback if needed.
- UI Optimization: Improve the Deployment UI with new features such as interactive visualizations of model predictions and user feedback integration.
- Scalability: Focus on optimizing the model's performance for large-scale data processing to handle a wider variety of car images in real-world applications.

3. Progress made in Reporting Week at an individual level and group level

3.1 Individual progress

Sakshi: Resolved repository configuration issues to ensure consistent deployments.

Bansil Patel & Meet Patel: Validated the model's robustness by testing on multiple datasets, confirming its generalization to different car damage types.

Harsh Mohile & Rachit Bhatt: Conducted extensive testing across staging and production environments, ensuring reliable performance.

3.2 Your team's progress

- Finalized the automated deployment pipeline with GitHub Actions, ensuring smooth deployments across all environments. Fully integrated the UI with the backend, ensuring seamless deployment in staging and production.

4. The areas/tasks you could not make progress and/or complete as scheduled or the difficulties encountered in this reporting week at individual level and group level.

4.1 Individual project blockers

Sakshi: Minor integration issues were resolved by refining repository configurations.
Bansil Patel & Meet Patel: Delays in backend-UI communication but resolved during testing.
Harsh Mohile & Rachit Bhatt: N/A.

4.2Your team's project blockers

- Multi-branch setup caused delays, resolved with configuration adjustments.
- Minor adjustments to align the UI with production configurations.

5. Tasks to be completed in the next week at the individual level and group level

5.1 Individual tasks

Sakshi, Bansil Patel & Meet Patel: N/A

Harsh Mohile & Rachit Bhatt: N/A

5.2 Your team's tasks

The project has been completed successfully.

6. Include the tasks from your sprint planning (GitHub/ZenHub) for the present period.

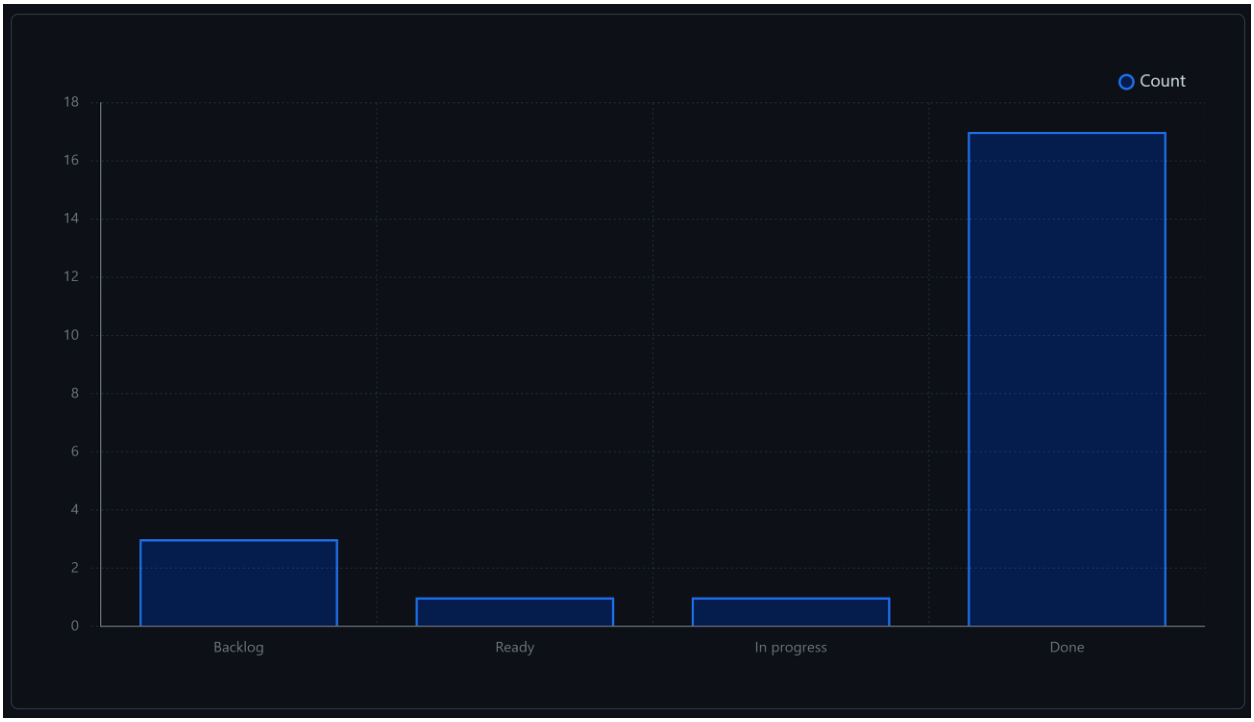

Figure 1: Weekly Activities

- In this week, there were a lot of issues resolved from the backlog, contributing to development.

7. **Include charts/graphs (e.g., burn down charts) from your project management tool (Github/Zenhub) that shows your progress for the period of this report.**

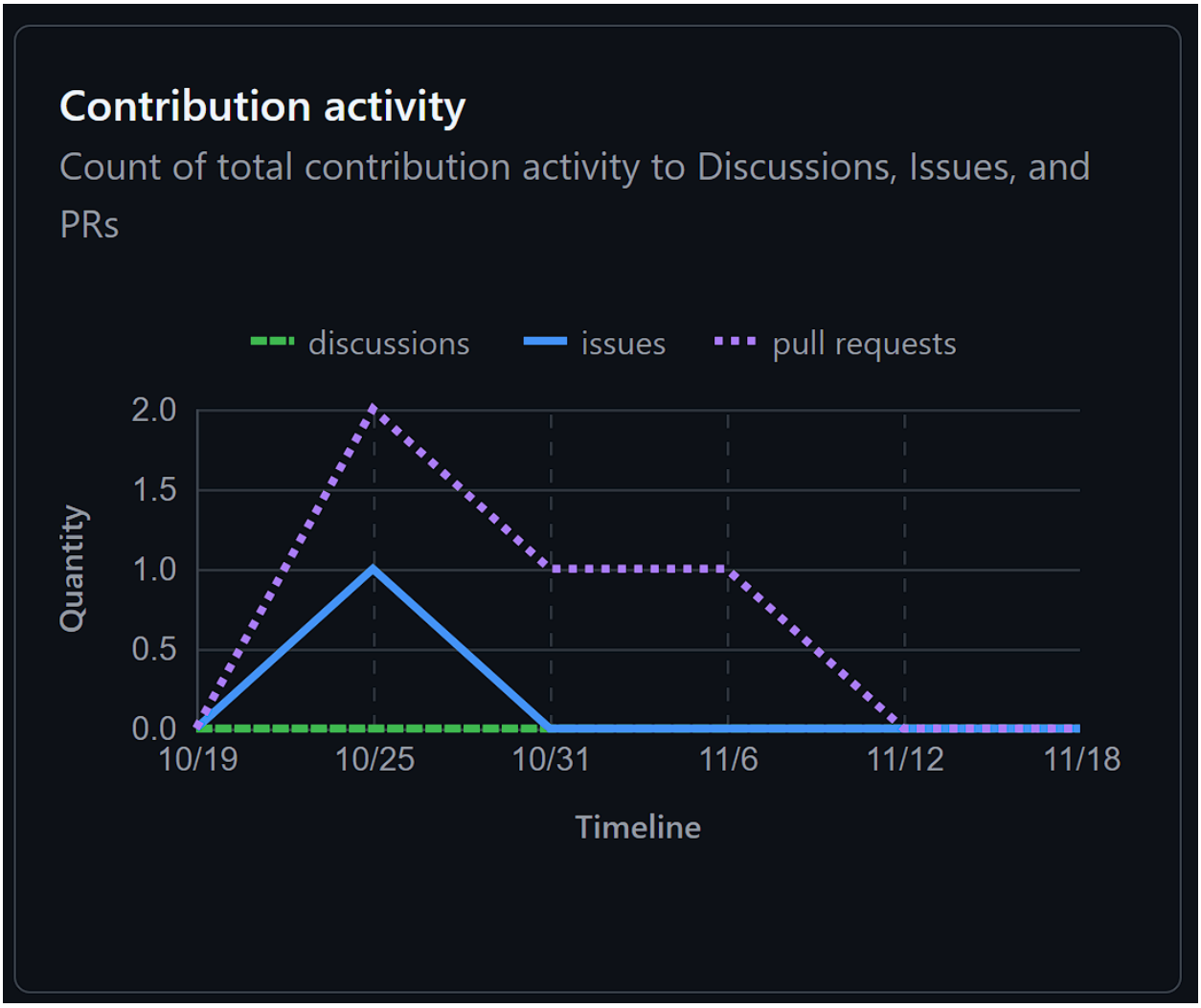

Figure 2: Contributors in the Week

- This week, we had a few PRs and issues resolved as compared to previous week, demonstrating team hard work and dedication.

8. **Include a note and address to your project GitHub with list of codes uploaded/updated on GitHub in this reporting week.**

GitHub: https://www.github.com/rachit-bhatt/Capstone-Project